Action semantics and timing provenance in robot imitation learning data

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Abstract

Robot imitation learning datasets can preserve numerical arrays while losing evidence needed to interpret acquisition and control semantics. We retrospectively audited 1,298 source and context episodes containing 664,369 rows stored on two robot installations, a paired 697-episode export, and 63 engineering recordings from four auxiliary installations. A separate comparator comprised 41 human-source BridgeData records. All paired exports preserved complete action, state, and row-timestamp arrays bit for bit, yet omitted 14 source timestamp columns, 363 diagnostic/reference columns, and original alignment summaries. Retained lineage linked exports to their source evidence. In 37 bottle-manipulation episodes, row timestamps followed the 30 Hz grid within 1.79 microseconds, while the episode-weighted mean proportion of rows with an inter-camera timestamp span exceeding one grid period was 33.14%. Simple state and affine controls remained predictive across three recording-group holdouts, whereas 32-state ridge failed despite small episode-held-out error. In contrast, a separately specified 16-state ridge remained predictive under reciprocal holdouts between two groups comprising 538 episodes on the second installation. These differing configurations constrain generalization of the failure. Stored approved-position references agreed closely with actions, although acquisition-time command causality remained unverified. Auxiliary records exposed unavailable action channels and missing diagnostic state fields; adjacent logs sometimes preserved timing evidence absent from arrays. The public comparator required a displacement-based action interpretation. These findings support separate reporting of numerical validity, evidence preservation, timing basis, action meaning, and outcome availability. They establish neither manipulation success nor a benefit from removing flagged data.

Keywords: imitation learning; robot demonstrations; data quality; temporal alignment; action semantics; behavioral cloning; reproducibility

1 Introduction

Learning from demonstration connects robot behavior to recorded observations and actions. Its practical appeal depends on the meaning of those records, not only their volume. A demonstration may contain a commanded joint target, a measured joint position, or a Cartesian displacement. These quantities can have similar numerical values while representing different supervision. The correspondence between the demonstrator, sensing system, and learner is therefore part of the learning problem (Argall et al., 2009).

Modern imitation learning methods make this issue more consequential. Action chunking and diffusion-based policies use temporally structured supervision to learn manipulation behavior (Zhao et al., 2023; Chi et al., 2023). Large datasets also combine tasks, collection interfaces, and robot embodiments (Walke et al., 2023; Open X-Embodiment Collaboration, 2024). A well-formed array alone does not identify the action convention or explain how observation streams were aligned. Evaluating data quality requires both numerical checks and evidence about the recording process.

This study addresses a narrower question than whether a demonstration is useful for successful imitation. We ask what an auditor can conclude from existing numerical trajectories, per-source timing provenance, and acquisition summaries. These records support reproducible consistency checks, but may lack the evidence needed to identify successful or expert demonstrations, or to evaluate learning utility.

Prior work has established that demonstration quality affects learning. Mandlekar et al. (2022) compared offline learning methods across human datasets with different collection conditions. Belkhale et al. (2023) analyzed quality through action divergence and transition diversity. Their formulation explains why a larger or more diverse dataset need not yield a better imitation policy. These studies concern the relationship between data and policy behavior. Our audit concerns the evidential prerequisites for interpreting data before that relationship is measured.

Quality also depends on the demonstrator and task. Sakr et al. (2022) connected demonstrated task performance with generalization in a user study. More recent work has examined demonstration consistency (Sakr et al., 2026). These approaches have access to outcomes or structured study conditions that our historical records do not retain. Their findings motivate careful quality assessment, but they do not justify deriving expert status from a trajectory's smoothness or numerical regularity alone.

Recent curation methods provide useful complementary perspectives. Kulkarni et al. (2026) proposed trajectory smoothness metrics, whereas Xu et al. (2026) used local gradient compatibility during policy adaptation. Both are currently cited as preprints. We do not reproduce or compare their learning benefits. Instead, we examine properties that can alter the interpretation of inputs to such methods, including channel correspondence, timing references, and missing source attribution.

Data standards and reproducible storage address another part of the problem. RLDS represents sequential decision data through explicit episode and step structure (Ramos et al., 2021). Mehak et al. (2024) discussed data quality standards for collaborative human-robot applications. These contributions motivate retaining acquisition context alongside arrays. Our analysis follows that direction through audits of two private recording archives and a separately sourced public representation.

Finally, offline prediction is a limited witness. Distribution shift can separate supervised imitation error from sequential performance (Ross et al., 2011). Evaluation leakage can also compromise machine learning conclusions (Kapoor and Narayanan, 2023). We therefore treat offline command regression as a diagnostic probe, include simple state-based controls, and compare episode-held-out, frame-random, and forward splits. We do not assume that a low regression error certifies demonstration quality.

The study asks three questions. First, which structural and semantic properties can be verified, and which survive export? Second, how do regular row timestamps, recorded source offsets, and phase-adjusted quality summaries differ? Third, what does offline predictability establish when simple controls and a distinct public action convention are considered?

The contribution is a paired empirical analysis of evidence preservation: we test whether a historical export retains the numerical data and the provenance needed to repeat an acquisition audit. Detailed source-timing checks, fixed prediction controls, and auxiliary engineering records identify properties that cannot be collapsed into one acceptance verdict. A second installation supplies reciprocal recording-group controls; external engineering provenance and a public semantic comparator test how these checks depend on the surrounding evidence bundle and action contract. This exploratory systems study does not introduce a new imitation algorithm or a validated quality classifier.

2 Materials and methods

2.1 Study design and recording provenance

We reanalyzed existing files without collecting new trajectories. The main private sample came from one K15-series humanoid installation. An initial inventory selected Bottle, Test, and ICL recording groups. A subsequent resource-driven extension recovered a larger historical group, denoted Source B, and its lineage-linked Export B. These are recording groups and paired representations, not randomized treatments or independent installations.

The primary group contained 37 episodes and 36,432 rows. Its stored task instruction described transferring a bottle between blue and orange coasters using the left hand, then returning the arm. Test contained eight episodes and 5,034 rows; ICL contained seven episodes and 6,935 rows. All three declared a 30 Hz sampling grid, the robot type k15-air-5, and LeRobot-compatible format version v3.0.

The format label does not identify the acquisition software revision. Source B contained 698 episodes and 418,903 rows. Export B selected 697 of these episodes, totaling 418,277 rows. Its task metadata described the bottle-transfer instruction, whereas the source used a generic collection label. Selected proxy clips supported a similar manipulation context; we did not infer task equivalence or success for every episode. Including the initial groups, the initial main-installation inventory contributed 750 source episodes and 467,304 rows. Derived export rows are not added to that denominator.

The numerical Parquet files, dataset metadata, episode metadata, and per-episode quality sidecars were copied to the analysis workstation. Initial reads were sequential and throttled; the larger extension used two bounded SFTP read channels with limited prefetch. File sizes and modification times were checked around numerical-file reads, and local SHA-256 digests were retained. These checks identify the analyzed files and detect changes observable through those attributes. They do not establish that the files were unmodified since acquisition or constitute a historical point-in-time snapshot.

Three Bottle head-camera proxies (first, shortest, and last episode) and three Source B proxies (episodes 0, 348, and 697) were inspected for recording context. The latter three decoded to their declared 939, 596, and 577 frames. This inspection supported a physical bottle-manipulation setting without assigning success labels or validating all image streams. The inspected raw-message directories for Bottle, Test, ICL and Source B were empty; source diagnostic fields were available in Source B, but demonstrator identities and a recording-time software revision remained unresolved.

The auxiliary inventory initially contained 41 nonempty historical engineering recordings from three installations: 39 acquisition/regression records, one capacity test, and one single-joint bench record, totaling 64,402 rows. Two empty metadata-only groups were inventoried separately. The fifth-round traversal recovered a fourth auxiliary archive, E4, including 12 legacy engineering groups (11,718 rows) with 7-, 8-, 14-, 16- or 18-channel action contracts. Six retained adjacent per-frame JSONL provenance. We tested structural pairing through row counts and contiguous frame indices and reconstructed offsets relative to stored grid timestamps; no authenticated acquisition-time payload link was available.

Further traversal of E4's formal recording root recovered two larger groups, E4-VR (344 episodes, 105,106 rows) and E4-Box (194 episodes, 56,171 rows), both with 16-channel action and state contracts. Task metadata described bimanual collection and placing an object into a box. Two smaller object/box context groups added five episodes and 34,522 rows. These source/context groups contribute 543 episodes and 195,799 rows; their labels identify recording contexts rather than verified successful demonstrations. E4 also contained ten additional engineering records (2,785 rows), giving 22 engineering records and 14,503 rows there. Together with the earlier three auxiliary archives, the engineering cohort contains 63 records and 78,905 rows. E4 contributes both source/context and engineering records; these categories do not constitute independent installations.

All numerical shards, episode metadata and available quality sidecars in these selected E4 groups were acquired. Multi-episode shards were split by episode index for analysis. A one-episode derived export and an empty metadata-only group were excluded from source totals. A partial local cache of E4-Box was reconciled to its canonical source and counted once. Six head-camera proxies, from the first, middle and last episodes of E4-VR and E4-Box, were fully decoded and visually sampled. They supported a physical gripper/object/box setting without certifying outcomes. Raw spool fragments were visible elsewhere on E4, but no authenticated mapping to the selected numerical episodes or independent command reference was recovered; they were not used as ground truth.

The broader online-fleet inventory recovered two further nonempty historical groups from the main installation, denoted Context C (three episodes, 407 rows) and Context D (two episodes, 859 rows), with stored task tags TEST and VR. All five head-camera proxies were inspected at four evenly spaced frames each and decoded to their declared lengths. They showed the physical table, robot hands and bottle/coaster scene, but did not establish equivalent tasks or successful outcomes. Three other retrieved groups were empty; four named candidate directories lacked dataset metadata. These cases remain in the coverage ledger without increasing the trajectory count. The additional groups extend within-installation context coverage. Combining the 755 main-installation episodes with the 543 E4 source/context episodes

gives 1,298 episodes and 664,369 rows stored on two installations (Table 1). Their availability-driven selection does not support a robot-population estimate.

A current local recorder source snapshot was retained to interpret field conventions. Its documented behavior agreed with several sidecar distinctions, including timing-basis-aware aggregation and camera phase adjustment. We did not establish that this source revision produced the historical files. Conclusions based directly on stored fields are separated from this corroborating implementation evidence.

2.2 Structural integrity and action contracts

Each numerical file was checked against its declared feature dimensions and episode length. We checked action and state finiteness, contiguous frame and global row indices within episodes, unique channel names, and consistent episode identifiers. Complete action-state episode arrays were fingerprinted to identify exact numerical duplicates. This procedure does not exclude partial overlap, common collection history, or near-duplicate behavior.

The primary data declared 26 action channels and 32 state channels. Its actions covered both seven-joint arms and two six-channel hands. State additionally included waist and head positions. Test and ICL declared 32 action and state channels, with waist and head channels appearing first. We joined actions to state by channel name, not by vector length or an assumed universal order. Position suffixes identify stored channels; units and calibration conventions were not independently certified. Physical interpretations therefore do not depend on converting these values to angular accuracy.

For the expanded E4 contracts, we also counted missing values by named field. Joint/gripper positions, diagnostic state fields, an absent state vector, and unavailable declared actions were distinguished rather than collapsed into an episode-wide quality score.

We inspected the stored control-source code and the corresponding measured-attribution flag. Their absence or sentinel values were treated as missing provenance. We did not infer a human operator from the robot-type label or from the presence of an action array. Exact equality between mapped state and action was reported descriptively, without using resemblance as a corruption criterion.

For the source-export audit, manifest selections and each export sidecar's derived-source identifier established the pairing. Every selected episode was untrimmed. We compared complete action, state, and row-timestamp byte representations after the declared float32 conversion, checked file SHA-256 values against the existing export manifest, and inventoried source timestamp, diagnostic/reference, alignment-summary, and control-summary retention. File hashes authenticate correspondence to that manifest, not the original collector. A separately stored deployment copy was compared through the same source identifiers, with state channels joined by name to test its 32-to-26-channel projection. Source and derived episodes were never treated as independent samples. An omitted source episode and missing sidecar were reported without inferring the selection reason.

Source B also retained per-joint telemetry labeled approved_pos and measured_pos. For the seven left-arm channels, we computed per-episode action distances and exact component agreement to each reference at the same stored row. These names and numerical relationships provide corroboration of the recorded representation. The two references are outputs of the same acquisition pipeline, not independent command or motion ground truth. Auxiliary records were checked against their own feature contracts; an intentionally absent state field was distinguished from nonfinite entries in a declared action vector.

2.3 Source timing and quality sidecars

The nominal grid instant for episode e and row i was reconstructed from its sidecar origin t0 and declared frequency f. For source s, the recorded signed offset was defined as follows.

$$\delta_{e,i,s} = \frac{t_{e,i,s} - t_{0e} - \frac{i \cdot 10^9}{f}}{10^9}$$

Here t denotes a recorded source timestamp in nanoseconds. We recomputed offsets from integer timestamps and compared them with the stored floating-point offsets. We also checked nonpositive timestamps, regressions, and repeated timestamps within each source and episode. A repeated timestamp indicates source-sample reuse in the stored representation. It does not establish a frozen sensor or a stationary robot.

For every episode and source, we summarized signed offsets and the 95th percentile of their absolute values. Exceedance proportions used half, one, and two nominal grid periods. These references express scale relative to the dataset cadence. They are diagnostic thresholds, not validated limits for safe control or successful imitation. Positive offsets indicate samples later than the offline grid instant; offline interpolation can legitimately use such samples.

Inter-camera timing was summarized through the source timestamp span in each row.

$$span_{e,i} = max_{c \in C}\ t_{e,i,c} - min_{c \in C}\ t_{e,i,c}$$

The camera set is denoted C, and the span was converted to milliseconds. It measures separation among recorded camera timestamps, not exposure synchronization accuracy. We had no independent clock or exposure measurement. Device-stamped and receive-stamped sources were retained as distinct categories, following their sidecars. The E4-VR and E4-Box groups each contain three cameras, whereas the initial main-installation groups contain four. Span comparisons retain this distinction and do not rank synchronization across installations.

Camera sidecars also contained phase offsets, phase-step information, and adjusted residual summaries. We retained these summaries separately from offsets relative to the original grid. The current recorder implementation centers camera offsets and can center separate plateaus after limited phase steps. Consequently, a small adjusted residual can coexist with a larger absolute grid offset. We examined overall and per-source verdicts independently rather than interpreting either as an all-purpose quality label.

We also reconstructed camera residuals by median centering, using separate phase plateaus when the sidecar indicated that adjustment. Overall verdicts were independently reconstructed from the stored per-source ratings and timing bases. These calculations check numerical compatibility with the reported conventions without identifying the historical software revision.

2.4 Offline command-regression controls

The original cross-validation comparisons used only the 37 Bottle episodes. Inputs were the 32 stored state channels; targets were the 26 stored action channels. Images, timestamps, episode identifiers, and task labels were excluded from model inputs. The probe asked how predictable recorded commands were from contemporaneous state. It was not a task policy evaluation.

Four fixed predictors were compared: the training-set mean action, state joined to action by channel name, an independent affine fit for each matched state-action channel, and multivariate ridge regression. Ridge used alpha = 1.0 and state standardization fitted only on training rows. Affine coefficients were estimated only from training data. No hyperparameter search used held-out observations.

The principal evaluation assigned complete episodes to five folds after a deterministic permutation with seed 20261019. Every episode received one set of held-out predictions. A frame-random five-fold comparison allowed neighboring rows from the same episode across training and evaluation. A forward sensitivity check trained on episodes 0–28 and evaluated episodes 29–36. The frame-random comparison is a diagnostic of splitting choices, not an independent replication. An additional five-fold check held out contiguous episode blocks and excluded their immediately adjacent episodes from training. A movement subset used left-arm state frame-difference speed above the training fold's 75th percentile; first rows were excluded from this subset. The fixed predictors and normalization were unchanged.

For a recording-group holdout, all 37 Bottle episodes trained the same four predictors, which were then evaluated on the 697 lineage-paired Source B episodes. Input and output channel names had to match exactly. No Source B observations were used to fit parameters, choose hyperparameters, or determine

normalizers. This tests transfer of the stored state-command mapping within one installation, not a task policy or a new robot.

The same Bottle-trained predictors were subsequently applied without refitting to Contexts C and D under a written extension plan. Inputs and targets were reordered using their declared channel names, allowing the 32-action Context C schema to be compared on the same seven left-arm targets. The original 32-state ridge and the two previously examined diagnostic variants were all retained. Because these historical groups were retrieved after the Source B failure was known, this is an additional retrospective check, not an independent prospective confirmation.

We emphasize mean absolute error over the seven left-arm channels, which correspond to the task instruction. We report errors in stored position units and normalize each channel by its training-set action standard deviation, floored at 0.001. The equation uses J for left-arm channels, T_e for evaluated rows, and σ_j for the training action standard deviation. Predicted and recorded actions are denoted â and a, respectively.

$$NMAE_e = \frac{1}{|T_e| \cdot |J|} \sum_{i \in T_e} \sum_{j \in J} \frac{|\hat{a}_{i,j} - a_{i,j}|}{max(\sigma_j, 0.001)}$$

The primary normalization does not combine hand and arm physical units. Supplementary analyses retain all 26 channels and examine standard-deviation floors of 0.01 and 0.1. Such checks matter because a small fixed discrepancy in a nearly constant channel can dominate a standardized aggregate.

For label sensitivity, we deliberately paired mapped state at row i with the recorded action at rows i, i+1, i+3, and i+10. All comparisons used the common interior rows 10 through n−11. These deterministic transformations operated only on local analysis arrays. They quantify sensitivity to label pairing and do not represent newly observed acquisition errors. No shifted version was designated a corrected dataset.

A separate E4 analysis used the two larger groups sharing an identical 16-state/16-action schema. It retained all 16 state inputs and evaluated the 14 named left- and right-arm joint targets, excluding gripper targets to avoid averaging arm and gripper units. The same four predictor families, training-only standardization, ridge alpha = 1.0, seed, and five whole-episode folds were used. Each full group also trained predictors evaluated on the other group. Exact whole-array overlap was checked before reciprocal transfer. Neither group selected a hyperparameter. This resource-driven extension was specified before reading its numerical payloads, after the original failure was known. It tests another observed contract, without isolating a causal effect of input dimension or providing robot-to-robot model transfer.

After observing the recording-group ridge failure, we performed two post hoc sensitivity analyses: a 0.001 floor on training input standard deviations, and a ridge model using only the seven left-arm state channels. Both retained alpha = 1.0, used Bottle for all fitting, and were evaluated under the original episode folds and the recording-group holdout. The variants were specified in their diagnostic script before execution, but this analysis was prompted by the failed holdout. We also inspected input range shifts and compared staged training indices with source alignment labels through export lineage. These diagnoses do not validate a deployment repair.

2.5 Public comparator and source verification

We downloaded the first official BridgeData TFDS 1.0.0 training shard, numbered 00000 of 01024, from the distribution linked by the BridgeData V2 project. Selection preceded numerical parsing. All 52 TFRecord records passed length and payload CRC32C checks. The shard contained 2,073 steps and included both human and scripted collection paths.

Source paths identified 11 scripted records, which were retained in the inventory but excluded from the human-demonstration comparison. The remaining 41 records contained 1,542 steps. Compound source-file and episode identifiers were used because the integer identifier alone was not globally unique. This is a convenience sample from one shard, not a prevalence estimate for BridgeData V2.

The public action convention was checked against official project documentation and pinned implementation snapshots. It comprises Cartesian and rotation increments with an absolute gripper

component, whereas the private channels describe joint and hand positions. The official training loader also offers reached-state relabeling for its first six action dimensions. We implemented the corresponding adjacent-state difference numerically, keeping position and rotation components separate. Recorded actions were paired with adjacent state differences over rows 0 through n−2, excluding the terminal action. Distances to zero action and absolute state used all n rows. These comparisons are representation diagnostics, not physical tracking errors or rankings of dataset quality.

BridgeData V2 is distributed under CC BY 4.0 according to its project page. Implementation code and dataset licensing were recorded separately. The public feature metadata and current converter snapshot were compared without assuming that the current converter exactly reproduced the historical shard.

2.6 Statistical analysis and reproducibility

The analysis was exploratory and retrospective. A written analysis specification was fixed after initial schema and offset inspection, before the reported regression comparisons. The path-based scripted-data exclusion and additional affine and normalization controls were documented as amendments. Emphasis on left-arm errors followed inspection of the all-channel normalization sensitivity; it was not a preregistered primary outcome. No robot-population hypothesis was tested. The fourth-round source/export audit, original auxiliary controls, contiguous-block check, and movement analysis followed resource-driven amendments fixed before their numerical summaries. The legacy E4 audit followed fixed scripts; its prose amendment was recorded after the first numerical summary. The later expanded-source and reciprocal-regression amendment preceded reading those numerical payloads. All extensions remained retrospective and resource-driven. Their purposes and exclusions are retained in the analysis specification.

Within a recording group, an episode was the summary unit. Frames and camera streams were not treated as independent experimental replicates. We report episode-weighted means, medians, interquartile ranges, and conditional percentile bootstrap intervals using 10,000 resamples and seed 20261019. Circular moving-block resampling of four consecutive episodes assessed sensitivity to local dependence. These intervals describe precision under episode or block exchangeability within the observed recording context. They do not estimate uncertainty across robots, sites, or demonstrators. The larger main-installation transfer and the two large E4 groups used a 32-episode block sensitivity interval; paired retention counts are exact descriptions of the inspected files and do not require sampling intervals.

Regression intervals summarize episode errors from fixed cross-validation predictions; models were not refitted inside the bootstrap. They therefore omit training-set and model-selection uncertainty. We report no p-values or claims of statistical superiority. Public representation comparisons are described separately, with no pooled private-public effect estimate. Software versions, split assignments, predictions, per-episode tables, figure sources, and file digests are retained in the reproducibility package.

3 Results

3.1 Structural checks passed while attribution remained unresolved

The initial 52 private numerical episodes matched their declared lengths, producing 48,401 rows in total (Table 1). Action and state values were finite, and inspected within-episode indices were contiguous. No complete action-state episode fingerprint was duplicated across the selected groups. These findings verify the inspected numerical exports; they do not validate uninspected original videos or rule out shared collection history.

Channel-name matching resolved the unequal primary dimensions without dropping the additional state channels. Treating equal vector dimensions as a prerequisite would have rejected the primary data despite its explicit mapping. Conversely, the identical dimensions of Test and ICL did not remove the need to inspect channel meaning and order.

The control-source code was −1 in every row of the initial groups, and their measured-attribution flags were zero. Quality sidecars described the trail as absent and the source as unattributed. We therefore

could not verify demonstrator identity, expertise, or control mode from the retained attribution fields. The primary task instruction and selected proxy clips provided context, but not episode-level outcome labels.

3.2 Regular grids did not imply simultaneous source timestamps

Row timestamps followed the declared 30 Hz grid within 1.79 microseconds across the initial three groups. Recomputed source offsets agreed with stored offsets within 3.19 nanoseconds. Within these three groups, no inspected source timestamps were nonpositive or regressed within an episode. Across source-episode tracks, repeated timestamps occurred 4, 2, and 29 times in Bottle, Test, and ICL, respectively. These are reuse counts across dependent streams, not counts of independently observed sensor failures.

Camera timestamps nevertheless differed within individual rows. In Bottle, the episode-weighted mean fraction of rows with a camera span above 33.333 ms was 33.14%, with a conditional bootstrap interval of 24.98–41.88% (Table 2). The four-episode block interval was 24.15–43.03%. The corresponding descriptive means were 57.34% for Test and 31.68% for ICL. Different tasks and collection conditions prevent interpreting these group differences as device-performance comparisons.

Absolute camera offsets and adjusted sidecar residuals described different quantities (Figure 1). For Bottle, the median episode-level absolute-offset 95th percentile was 23.54 ms for the head camera and 25.68 ms for the chest camera. Median adjusted sidecar values were 9.81 and 12.94 ms, respectively. All four primary camera tracks were rated acceptable in every episode. These observations show that acceptable residual timing does not imply zero phase offset or simultaneous samples.

Across all 208 camera-episode tracks, reconstructed residual percentiles differed from stored sidecar values by at most 0.0432 ms. Timing-basis-aware reconstruction matched all 52 overall verdicts. This numerical agreement corroborates the distinction between reference quantities while leaving the exact historical implementation unresolved.

The overall alignment rating was acceptable in 35 Bottle episodes, all eight Test episodes, and five ICL episodes. However, every command-stream sidecar was degraded. Command streams used receive-time provenance, whereas the overall verdict followed the device-stamped sources when available. The stored distinctions and current implementation explain why these verdicts can coexist. They do not independently demonstrate command corruption or certify camera-command synchronization.

3.3 Simple controls explained much of offline predictability

Episode-held-out ridge regression produced a mean left-arm normalized error of 0.00635, compared with 0.00882 for mapped state and 0.00724 for channel-wise affine prediction (Table 3). The training-mean baseline yielded 0.81234. Native left-arm errors were 0.00147, 0.00224, and 0.00170 for ridge, mapped state, and affine prediction, respectively. These results indicate close contemporaneous state-command correspondence in the stored data.

Frame-random ridge evaluation produced a normalized error of 0.00613. Forward evaluation on the last eight episodes yielded 0.00626. The observed difference between frame-random and episode-held-out evaluation was small, so these data do not support a large leakage-inflation claim. Neither evaluation separated robots, recording installations, or demonstrators. Contiguous-block evaluation with an adjacent-episode embargo gave ridge NMAE 0.00658 (conditional interval 0.00607–0.00726). The movement subset retained 9,141 rows across all 37 episodes and yielded 0.00878, versus 0.01074 for mapped state and 0.00954 for affine prediction. Thus, the small error did not depend entirely on stationary rows or adjacent-episode training. Figure 2 summarizes the episode distributions and label-sensitivity check.

Deliberately advancing action labels increased the mapped-state left-arm error. Using the same interior rows, the episode-weighted native error rose from 0.00224 at zero shift to 0.00513, 0.01148, and 0.03383 at shifts of 1, 3, and 10 rows. This establishes sensitivity of this probe to the imposed pairing. It does not identify an optimal recording delay or prove that the original alignment was physically correct.

All-channel standardized errors were much more sensitive to normalization choices than the active-arm summaries. Nearly constant channels amplified small offsets under the 0.001 floor. The supplementary

results retain this unfavorable sensitivity rather than presenting the largest apparent regression gain as the principal result.

3.4 Numerical preservation did not preserve the acquisition audit

Source B's 698 numerical episodes matched 418,903 declared rows; its action and state values were finite. The export selected 697 complete episodes and omitted source episode 79 (626 rows). That source episode also lacked a quality sidecar. The selection manifest did not establish why it was omitted. Complete source episode fingerprints had 0 internal duplicates and 0 matches to the initial 52 episodes; this does not exclude near-duplicate behavior or shared collection context.

All 697 source-export pairs preserved action, state, and row timestamps bit for bit (Table 4). Nevertheless, every export omitted the 14 source timestamp columns and 363 diagnostic/reference columns present in each source episode. These are schema-retention counts: 225 reference fields were entirely finite, 96 entirely nonfinite, and 42 had mixed coverage; these counts do not imply 363 independent informative signals. Original alignment summaries were absent from all 697 export sidecars; control summaries and explicit source identifiers remained. The manifest declared with_provenance=false, and sidecars described their limited scope. This was a documented representation choice, not an inferred recorder malfunction. Without the original source, source-time alignment could not be recomputed from these exports. A second, 26-state deployment copy preserved actions, row timestamps and the named state projection bit for bit in all 697 episodes. Its formal selection of 676 episodes (399,088 rows) matched the staged model indices and the source acceptable-alignment subset through lineage. This verifies the inspected representation and selection correspondence; the configured training-root path differed, so actual training-payload identity remains unverified (Section S9).

Across all Source B episodes, the mean fraction of rows with an inter-camera span exceeding one grid period was 36.23%. Among the 697 available source reports, 676 were alignment-acceptable. Attribution columns remained unattributed, although separate telemetry contained diagnostic source codes. The left-arm episode-mean action distance to the approved-position reference was 0.000085, versus 0.003230 to the measured-position reference (Figure 3A). This corroborates command-related numerical content without authenticating its historical production chain.

3.5 Recording-group holdouts exposed multivariate prediction failures

The recording-group holdout yielded left-arm native errors of 0.00220, 0.00182, and 21.91924 for mapped state, affine, and ridge prediction, respectively; the training-mean error was 0.21154 (Figure 3B). State and affine correspondence persisted, whereas multivariate ridge failed under a recording-group shift concealed by within-group folds. No held-out recording-group data fitted these models. None of these comparisons establish visual control or manipulation success.

The failure persisted in the 676 alignment-acceptable Source B episodes. Inspection found a training standard deviation of 1.44×10^{-6} for the lower-waist pitch channel, compared with 0.0241 in the held-out group; several inactive-arm and head channels also shifted. Flooring input scales at 0.001 reduced the recording-group native error to 1.86714 but did not recover the original accuracy. Restricting ridge to left-arm state gave 0.00172 across groups and 0.00152 under episode cross-validation. These post hoc controls implicate sensitivity to the broader input distribution without identifying a unique physical cause or an optimal feature policy.

The failure also occurred in both additional context groups. Unchanged 32-state ridge native MAE was 665.91562 in Context C and 69.53270 in Context D, compared with 0.00253 and 0.00292 for mapped state (Figure 3B). All five episodes had finite arrays and acceptable alignment summaries. The earlier input-scale-floor variant still yielded 7.106997 and 9.165710, whereas the already specified left-arm-only diagnostic yielded 0.002834 and 0.002906 (Table S6). These are offline prediction errors in stored units, not observed physical deviations. The two- and three-episode groups provide additional examples of the same vulnerability; they cannot establish a general feature-selection remedy.

3.6 The second installation did not reproduce the severe ridge failure

E4-VR and E4-Box contributed 538 episodes and 161,277 rows with finite 16-channel action and state arrays. No complete action-state fingerprint was duplicated within or across these groups. The fixed 16-state ridge achieved native MAEs, averaged over the 14 arm targets, of 0.000818 and 0.000877 under episode-held-out evaluation. Reciprocal transfer gave 0.001284 for VR-to-Box and 0.001065 for Box-to-VR, compared with mapped-state errors of 0.001494 and 0.001504 (Table 5). The training-mean control remained substantially farther from recorded commands. Thus, these recording-group shifts did not reproduce the severe failure of the primary 32-state model. Different inputs, target sets, tasks, collection conditions and sample sizes prevent attributing this contrast to dimensionality alone.

The two groups also retained distinct timing summaries. Their mean camera-span exceedance proportions were 13.14% and 11.35%, with three cameras per group (Table 2). Source-offset reconstruction agreed within 3.20 ns and row-grid residuals were below 0.90 microseconds. Overall alignment was acceptable in 343/344 VR episodes and 138/194 Box episodes. The Box sidecars lacked a length-rating field; its absence was not converted to a failed rating. Stored action-source annotations were retained as metadata, without authenticating operator identity or acquisition-time command causality. These checks describe the stored representation and do not establish superior physical synchronization relative to the four-camera primary records.

3.7 Engineering records separated validity from provenance availability

Auxiliary engineering records exposed a different audit boundary. Three 30 Hz loop records contained NaNs throughout the left-gripper action channel, which had no matching state channel. A camera-rehearsal record declared no state vector and had NaNs in all 22 action channels, despite an acceptable alignment verdict. Its camera-only context makes it unsuitable as a joint-action demonstration without additional information. The other 37 records in the initial three auxiliary archives had finite action vectors. These are configuration-specific observations, not estimates of demonstration-defect prevalence.

The 12 legacy E4 arrays were finite and had no embedded source-timestamp columns or original quality sidecars. Adjacent JSONL files nevertheless supplied 6,808 structurally paired rows for six groups (Table S7). Of 27 source-episode timing tracks, 23 were fully populated and four action tracks were partial. Recomputed offsets matched stored values on jointly available rows. Thus, missing columns did not imply that timing evidence was absent from the surrounding bundle. One group had exactly equal action and state arrays; a retained producer script described passive observation-to-action echo. Without an authenticated historical build, that script corroborates a possible intended convention rather than establishing the recording-time implementation.

Additional E4 configurations clarified the meaning of nonfinite entries. Two restart-verification records contained NaNs in six declared waist/head action channels across all 281 rows (1,686 cells); their 16 stored joint/gripper state channels remained finite. Three task-context records and six engineering records used a 121-channel state vector with missing values in 14 named teleoperation diagnostic fields. All 16 joint/gripper position fields in those vectors were finite. Such diagnostic missingness affects a consumer requiring those fields but does not by itself invalidate the position targets. The audit therefore reports field-level coverage separately from an all-fields-finite verdict (Table S8).

3.8 The public comparator required a different semantic interpretation

The 41 human-source BridgeData records passed the inspected numerical shape and episode-boundary checks. Every record began with first-six-component motion values below 10^{-5} in magnitude. This boundary pattern was retained, rather than automatically classified as a missing command. The public format's seven-dimensional state and action arrays did not imply identical physical quantities.

For the three position components, the mean episode-level numerical distance between recorded actions and reached-state differences was 0.00605 in stored units. The corresponding distance to zero action was 0.00931; the distance to absolute state was 0.15189. These values reflect distinct representations (Figure 4). They are not commensurate with private joint-position errors and do not measure task quality.

The downloaded public feature metadata declared seven-element arrays but described ten components for both state and action. The current pinned converter instead documented seven end-effector and gripper components. This discrepancy is present in the inspected metadata snapshot and is not evidence that the numerical trajectories themselves are malformed. It illustrates why checking array dimensions against prose and authoritative implementation is useful even in an established public distribution.

4 Discussion

The paired audit establishes that numerical preservation and evidence preservation are distinct properties. An export retained complete supervised-learning arrays while deliberately omitting the fields needed to repeat source-timing and diagnostic checks. Recovering the source made the difference measurable. The detailed timing analysis, stable state-based prediction, and configuration-specific engineering cases explain why one favorable verdict cannot certify every property of a training record.

4.1 Timing quality depends on the reference quantity

A recorder must map asynchronous sources onto a training representation. A regular grid simplifies indexing, but it does not erase differences among source times. Phase adjustment can assess the stability of a camera's own cadence while preserving an offset from the common grid. These are compatible engineering objectives. Treating the adjusted residual as a measurement of physical simultaneity would change the scientific claim without changing the underlying data.

The camera-span observations also have alternative explanations. Fixed phase relationships, device clock conventions, acquisition buffering, and processing choices can all separate stored timestamps. We lacked independent exposure timing and historical clock-calibration evidence, so the audit cannot isolate these mechanisms. The appropriate conclusion is that the retained representation exposes temporal separation whose physical and learning consequences remain unmeasured.

The command verdicts reinforce this interpretation. Receive-time timing can depend on host scheduling as well as source behavior. Excluding such streams from a device-timing aggregate can be a deliberate design choice. However, an analyst using that aggregate to select training data still needs to inspect command provenance separately. A summary label is informative only within the scope of its aggregation rule.

4.2 Predictable labels are an incomplete quality witness

The simple state and affine controls explain why a small regression loss should be interpreted cautiously. Absolute targets can remain close to measured positions during smooth tracking, and channel calibration can create stable offsets. Other possibilities include shared processing or feedback-derived labels. Source B's approved-position telemetry supports a command-related interpretation more directly than state resemblance alone. However, a common pipeline could create both recorded fields, so this agreement does not authenticate a historical causal chain. Raw messages and a verified recording revision are still needed to distinguish those explanations; action echo or label leakage is not established.

This boundary is consistent with the distinction between supervised imitation error and sequential distribution shift (Ross et al., 2011). It also complements data-quality formulations that depend on policy behavior and transition properties (Belkhale et al., 2023). Our regression probe establishes a reproducible property of the stored mapping. It does not establish causal action availability, visual grounding, recovery behavior, or manipulation success.

The recording-group failure adds a separate limitation of favorable within-group loss. A multivariate model can use nearly constant or correlated state channels that change in another collection, even on the same robot. The stable affine control and task-restricted sensitivity check make that alternative visible. Recurrence in two later historical context groups shows that the failure was not confined to Source B. The E4 reciprocal holdouts supply a necessary counterpoint: another contract on a second installation retained small ridge errors. Together, the results localize the observed vulnerability to particular tested mappings and distribution shifts rather than ridge regression or robot demonstrations in general. The left-arm

restriction and input-scale floor remain post hoc diagnostics; the E4 contrast is not a controlled validation of either remedy.

The normalization analysis adds a practical qualification. Aggregating standardized errors across inactive and active joints can exaggerate the importance of small fixed discrepancies. Native-unit summaries and task-relevant channel groups make the result easier to interpret. They still require an explicit action convention, especially when comparing absolute joint targets with Cartesian increments.

4.3 Implications for demonstration audits

The public comparator shows that an audit needs a contract before it can interpret numerical resemblance. Reached-state differences are an available relabeling convention in the official BridgeData training code. Their difference from recorded commands is therefore not automatically a defect. Likewise, a near-zero initial motion vector can follow a boundary convention. Generic thresholds applied without these distinctions risk conflating representation choices with acquisition failures.

The paired export and E4 logs identify a preservation requirement: compact arrays can support an audit when their external provenance remains accessible and securely linked. Explicit source identifiers are useful only while the corresponding artifacts remain available. E4 also shows why an auditor should search the surrounding bundle before declaring provenance unavailable; matching row indices alone does not authenticate that link. Selective omission is legitimate, but the release should state which checks can no longer be performed from the distributed representation.

A practical audit should retain separate records of numerical validity, channel meaning, source timing, control attribution, and outcome availability. This organization extends standards-oriented arguments for retaining acquisition context (Mehak et al., 2024). It also supplies interpretable inputs to later curation methods, without assuming that any particular audit flag predicts rollout utility. Smoothness or learned curation scores can then be studied against outcomes under a verified contract, rather than used to fill provenance gaps retrospectively.

Two further analyses would resolve the most consequential uncertainties. A verified raw-message replay could establish whether action labels came from commands, feedback, or transformed sources at acquisition time. An independent evaluation with authorized rollouts could then compare unchanged data, random retention, and prespecified audit-based retention. That comparison would test learning utility while controlling dataset size. Neither result is supplied by the current retrospective analysis.

4.4 Limitations

The learning-related analyses cover archives stored on two installations, with different contracts and recording contexts. They do not constitute randomized, task-equivalent robot or site replication. The four auxiliary archives, one shared with the second source archive, contribute engineering checks under different configurations. Additional episodes improve within-context coverage without identifying independent operators or collection sites. Block and recording-group holdouts constrain some dependence explanations but cannot estimate between-site learning performance. The fleet inventory has explicit access, permission, depth and scope limits; storage availability also determined source selection.

The recording provenance is incomplete. Robot-type metadata and selected clips support a physical-robot context, but operator identity, expertise, control mode, and per-episode success remain unverified. The available format version is not a recording-time software version. Current source snapshots clarify plausible semantics but do not authenticate the historical pipeline.

The historical export was selected by its original producer, and its selection criteria were not independently recovered. Paired equality validates the inspected transformation, not the quality of excluded or retained demonstrations. Our timing measures depend on retained timestamp conventions. They do not provide external synchronization truth, end-to-end sensor latency, or control causality. Full raw streams and most videos were not analyzed. A numerical pass therefore cannot be extended to image decoding, visual quality, or every step of acquisition and export.

The offline predictors use state alone and cannot resolve perceptual ambiguity. Their evaluation omits closed-loop error accumulation and generalization to independent robots or tasks. The deliberately shifted labels test a mathematical sensitivity, not the effectiveness of a proposed repair. No curation policy was validated against success labels.

Finally, the public comparison is restricted to one convenience shard and a provenance-based subset. The metadata discrepancy should not be generalized to all BridgeData versions or other public datasets. Its value here is a reproducible counterexample to relying on numerical dimensions without checking the associated contract.

5 Conclusion

An audit of existing robot imitation learning records can establish numerical consistency while leaving acquisition and learning utility unresolved. In 697 verified source-export pairs, complete learning arrays survived while acquisition timestamps and diagnostic references were intentionally omitted. Regular grids, adjusted timing verdicts, and low regression errors described additional, different properties. Channel-aware controls, reference comparisons, auxiliary engineering records, and a public representation comparison made those distinctions measurable. Three main-installation recording-group holdouts exposed multivariate prediction failure concealed by within-group folds, while reciprocal holdouts under a different contract on the second installation did not. This contrast limits the generality of the failure. The resulting evidence supports explicit reporting of action semantics and timing provenance, with separate treatment of missing attribution and outcome labels. Claims that such checks improve policy success require an additional evaluation that was not performed here.

Data availability statement

The public comparator is available through the official BridgeData V2 project and its TFDS distribution. Derived tables, analysis scripts, split assignments, and figure sources are included in the accompanying reproducibility materials. The underlying industrial recordings are not deposited in a public repository with this manuscript draft. Their redistribution or mediated access requires data-owner authorization. [Authors must finalize the repository identifier and justified access conditions before submission.]

Ethics statement

This study reanalyzed existing robot recording artifacts and did not recruit participants or initiate new physical robot experiments. The retained control provenance does not identify individual demonstrators. [Authors must confirm the applicability of institutional review, consent, and secondary-use requirements for the historical recordings.]

Author contributions

[Authors must assign verified CRediT roles and approve the final author list.]

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[Authors must provide verified funding information or confirm the applicable no-funding statement.]

Conflict of interest

[Authors must declare relevant employment, commercial relationships, and other competing interests.]

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Supplementary material

Supplementary material includes extended audit definitions, normalization and dependence checks, source-data tables, and the reproducibility instructions.

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Tables

Table 1. Inspected records and evidential scope. Bottle, Test, ICL, Source B and Contexts C/D share one installation; the E4 source/context groups share a second. The source/context total is 1,298 episodes and 664,369 rows. Derived exports are excluded from that total. Engineering controls cover four auxiliary archives, including E4; they are not task-equivalent replications. Bridge is a separate convenience sample.

Sample	Episodes	Rows	Action/state dimensions	Role
Bottle	37	36,432	26 / 32	Primary audit and offline probes
Test	8	5,034	32 / 32	Within-installation context
ICL	7	6,935	32 / 32	Within-installation context
Context C	3	407	32 / 32	Additional within-installation context
Context D	2	859	26 / 32	Additional within-installation context
Source B	698	418,903	26 / 32	Source audit; 697 paired exports
Export B (derived)	697	418,277	26 / 32	Paired representation; not added to source total
Deployment B (derived)	697	418,277	26 / 26	Named state projection; not added to source total
E4-VR	344	105,106	16 / 16	Second-installation source and fixed probes
E4-Box	194	56,171	16 / 16	Second-installation source and fixed probes
E4 small contexts	5	34,522	16 / 16 or 121	Source/context audit; no outcome labels
Engineering E1–E4	63	78,905	Configuration-specific	Auxiliary audit controls; four archives
Bridge human subset	41	1,542	7 / 7	Public semantic comparator
Bridge scripted subset	11	531	7 / 7	Inventoried, excluded from human comparison

Table 2. Camera timestamp spans and overall sidecar verdicts. Exceedance is the per-episode fraction of rows whose largest minus smallest camera timestamp exceeds 33.333 ms, averaged equally over episodes. Intervals are conditional bootstrap precision summaries, not population intervals. Circular blocks contain four episodes for the initial groups and 32 for E4. Camera counts and collection conditions differ, preventing device-performance ranking.

Group	n	Cameras	Mean exceedance (%)	Episode interval (%)	Block interval (%)	Overall ok
Bottle	37	4	33.14	24.98–41.88	24.15–43.03	35/37
Test	8	4	57.34	32.84–79.50	47.76–66.91	8/8
ICL	7	4	31.68	12.76–47.88	17.97–45.38	5/7
E4-VR	344	3	13.14	10.44–16.01	10.92–15.56	343/344
E4-Box	194	3	11.35	8.08–15.10	8.82–13.90	138/194

Table 3. Offline left-arm command-regression errors. Means give equal weight to each evaluated episode. NMAE uses training action standard deviations with a 0.001 floor. Native errors use stored position units. Bootstrap intervals condition on fixed predictions and do not include model refitting uncertainty.

Predictor / split	n	Mean NMAE	95% conditional interval	Mean native MAE
Mean / episode	37	0.81234	0.77965–0.85286	0.22940
State / episode	37	0.00882	0.00855–0.00916	0.00224
Affine / episode	37	0.00724	0.00699–0.00754	0.00170
Ridge / episode	37	0.00635	0.00598–0.00684	0.00147
Ridge / frame	37	0.00613	0.00588–0.00645	0.00142
Ridge / forward	8	0.00626	0.00595–0.00655	0.00146

Table 4. Evidence retention in the 697 complete source-export pairs. All acquired export files matched their existing manifest hashes. Source episode 79 (626 rows) was not selected and lacked a quality sidecar; its exclusion reason was not established. Column counts are per episode. Equality describes the stored arrays and does not certify acquisition history.

Property	**Paired source**	**Export**
Action / state / row timestamps	697 complete episodes	697 bit-identical pairs
Source timestamp columns	14	0
Diagnostic/reference columns	363	0
Original alignment summaries	697	0
Control summaries	697	697
Explicit derived-source identifiers	Not applicable	697

Table 5. Second-installation fixed prediction controls. Inputs comprise 16 state channels and targets 14 named arm joints; gripper targets are excluded. Entries are episode-weighted mean native MAEs. Episode evaluation uses five whole-episode folds; reciprocal transfers fit one full group and evaluate the other. All fits and standardization use training rows only. These contracts differ from the primary 32-state/left-arm analysis, so their errors are not a controlled comparison of input dimensions. Conditional intervals and normalization sensitivity are retained in Table S9 and machine-readable results.

Training / evaluation	**n**	**Mean**	**State**	**Affine**	**Ridge16**
VR / episode	344	0.099682	0.001504	0.000951	0.000818
Box / episode	194	0.138756	0.001494	0.001022	0.000877
VR → Box	194	0.132608	0.001494	0.001176	0.001284
Box → VR	344	0.126028	0.001504	0.001052	0.001065

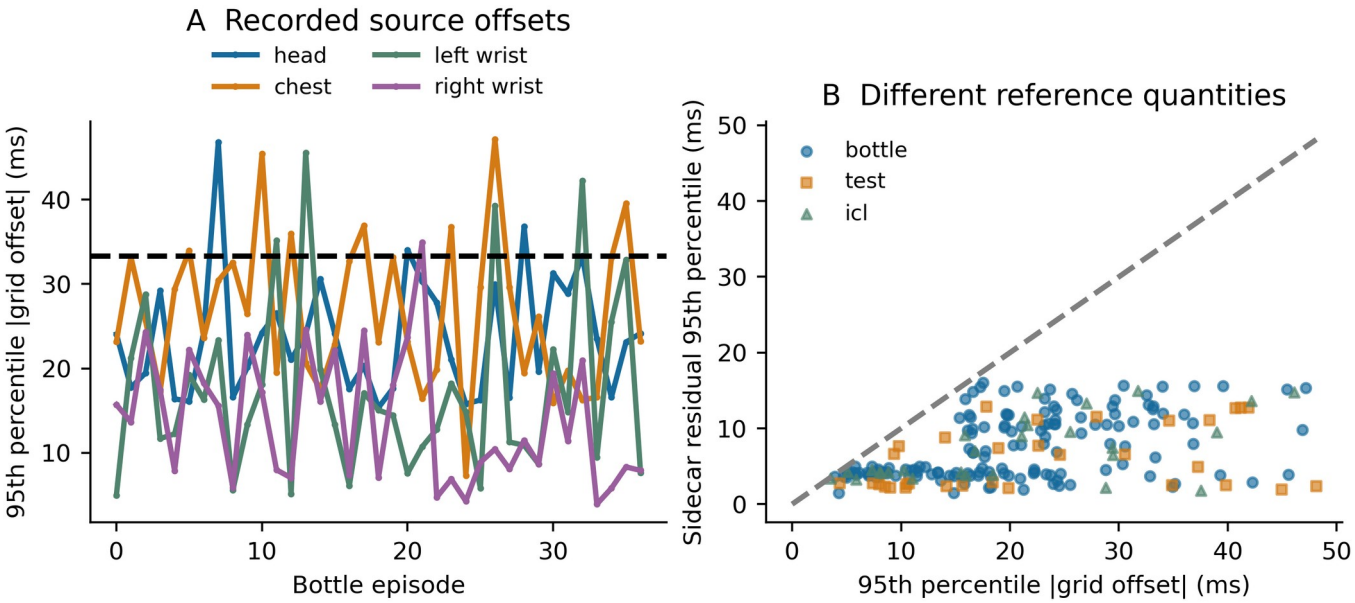


Figure 1. Grid-relative offsets and phase-adjusted camera summaries. (A) Episode-level 95th percentiles of absolute recorded offsets in the 37 Bottle episodes; the dashed line marks one 30 Hz period. (B) Absolute grid-offset percentiles plotted against stored sidecar residual percentiles, with one point per camera and episode. The diagonal indicates numerical equality. Residual phase adjustment changes the reference quantity; low residuals do not establish physical simultaneity.

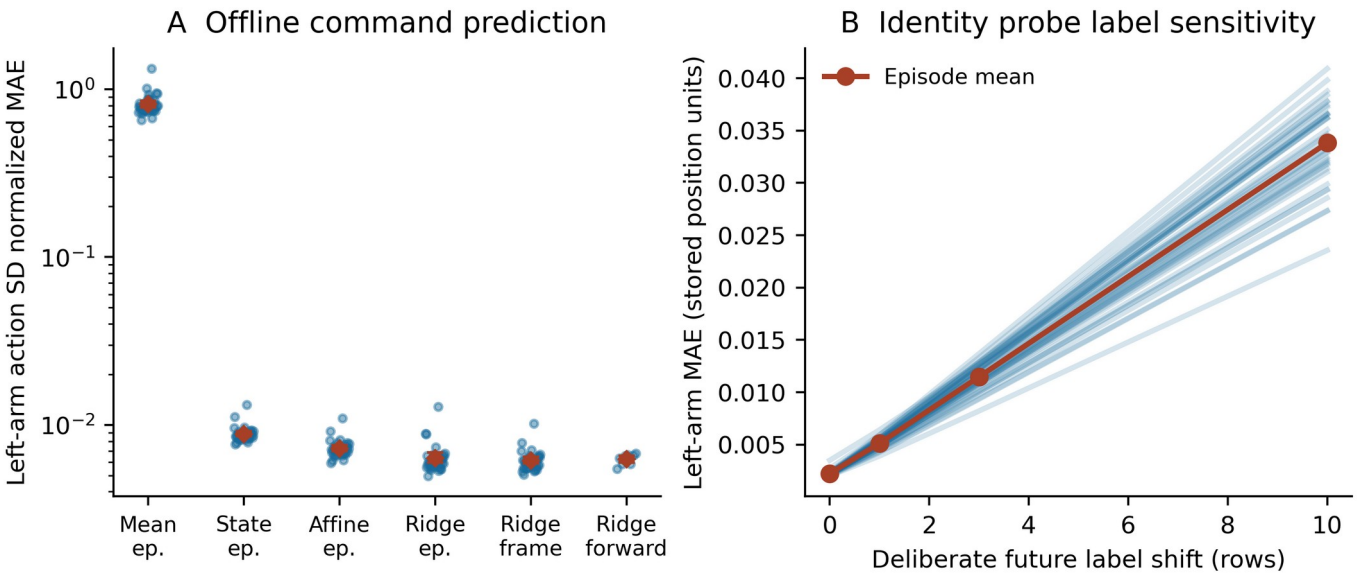


Figure 2. Offline predictability and imposed label shifts. (A) Left-arm NMAE for fixed predictors and split protocols. Small points are episode errors; diamonds show episode means with conditional 95% percentile intervals. The vertical scale is logarithmic. Forward evaluation uses eight episodes; other comparisons use 37. (B) Mapped-state native error after deliberately advancing action labels, using identical interior rows. Pale lines are episodes and the dark line is their mean. These imposed shifts test label sensitivity rather than a recording repair.

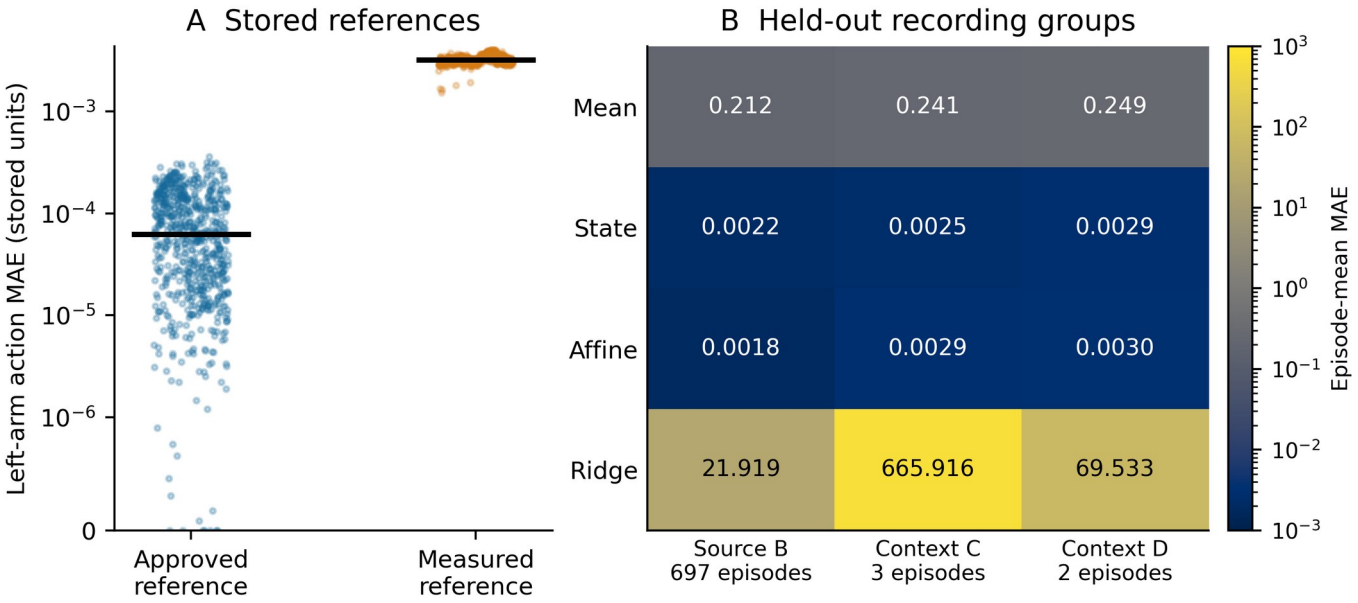


Figure 3. Reference correspondence and three recording-group holdouts. (A) Left-arm action MAE to approved-position and measured-position telemetry in 698 Source B episodes; points are episodes and black lines mark medians. The symmetric logarithmic scale is linear below 10^{-6}. These references share an acquisition pipeline. (B) Episode-weighted mean native MAE for four unchanged predictors trained on 37 Bottle episodes. Held-out groups comprise 697 Source B episodes, three Context C episodes and two Context D episodes. Cell colors use a logarithmic scale; numbers give the mean in stored position units. Groups are reported separately and do not represent independent installations. No held-out observations were used to fit the models or normalizers.

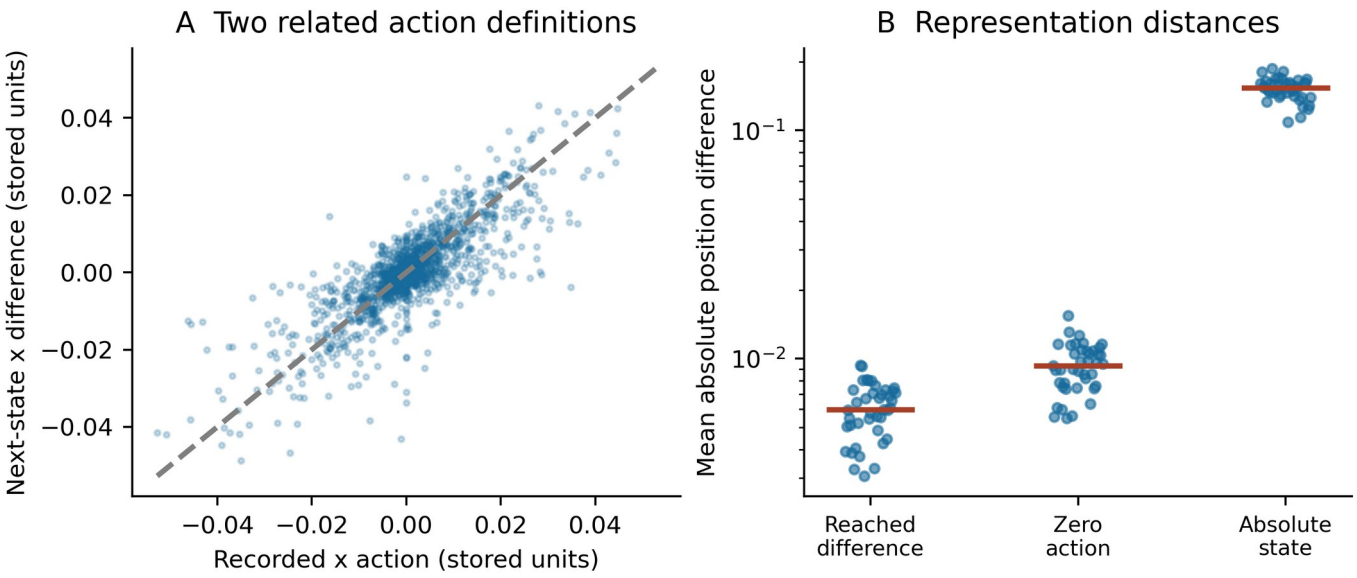


Figure 4. Public action-contract diagnostics for 41 human-source BridgeData records. (A) Recorded x displacement against the adjacent reached-state x difference, with an equality reference. (B) Per-record position-component numerical distances to reached differences, zero action and absolute state. Horizontal marks indicate medians. Representations are intentionally different; distances are not physical tracking errors or dataset-quality rankings. Scripted records are excluded.